EASY · ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

04

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN ADVANCED MATH

$$x - 2 = 9$$

What is one possible solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2 ADVANCED MATH

$$x = 8$$

$$y = x^2 + 8$$

The graphs of the equations in the given system of equations intersect at the point x, y in the xy-plane. What is the value of y?

A. 8

B. 24

C. 64

D. 72

Q3 ADVANCED MATH

$$\frac{x^2}{25} = 36$$

What is a solution to the given equation?

A. 6

B. 30

C. 450

D. 900

$$8x^2 - 40 = 32$$

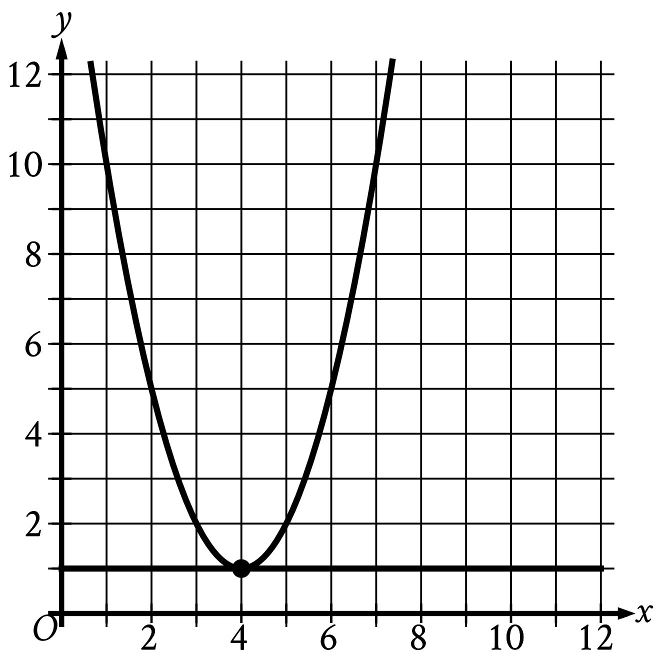
What is the positive solution to the given equation?

A. 3

B. 4

C. 9

D. 72



- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (4 comma 1).
  - The parabola passes through the following points:
    - (3 comma 2)
    - (4 comma 1)
    - (5 comma 2)
- For the line in the system:
  - The line is horizontal.
  - The line passes through the following points:
    - (0 comma 1)
    - (4 comma 1)

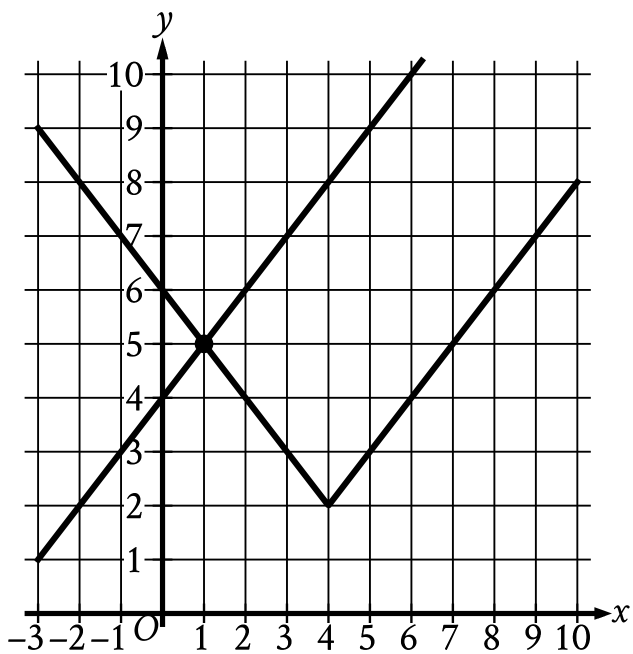
The graph of a system of a linear and a quadratic equation is shown. What is the solution x, y to this system?

A. $0, 0$

B. $-4, 1$

C. $4, -1$

D. $4, 1$



- For the absolute value function:
    - Moving from left to right:
        - The function slants sharply down to the point (4 comma 2).
        - The function then slants sharply up.
    - The function passes through the following points:
        - (1 comma 5)
        - (4 comma 2)
        - (6 comma 4)
- For the linear function:
    - The function slants sharply up from left to right.
    - The function passes through the following points:
        - (0 comma 4)
        - (1 comma 5)

The graph of a system of an absolute value function and a linear function is shown. What is the solution x, y to this system of two equations?

A. -1, 5

B. 0, 4

C. 1, 5

D. 4, 2

Q7 ADVANCED MATH

$$y - 57 = px$$

The given equation relates the positive numbers p, x, and y. Which equation correctly expresses y in terms of p and x?

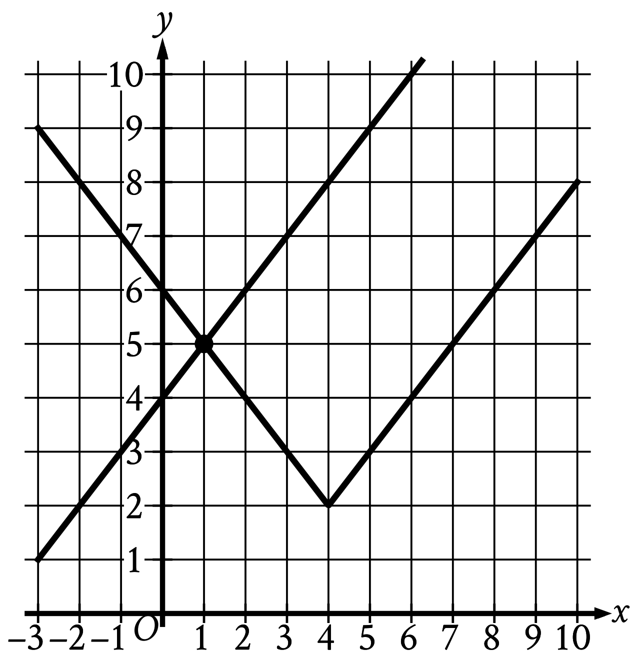
A. $y = 57x + p$

B. $y = px + 57$

C. $y = 57px$

D. $y = \frac{px}{57}$

Q8



- For the absolute value function:
  - Moving from left to right:
    - The function slants sharply down to the point (4 comma 2).
    - The function then slants sharply up.
  - The function passes through the following points:
    - (1 comma 5)
    - (4 comma 2)
    - (6 comma 4)
- For the linear function:
  - The function slants sharply up from left to right.
  - The function passes through the following points:
    - (0 comma 4)
    - (1 comma 5)

The graph of a system of an absolute value function and a linear function is shown. What is the solution x, y to this system of two equations?

A. -1, 5

B. 0, 4

C. 1, 5

D. 4, 2

Q9

ADVANCED MATH

$$x^2 = 64$$

Which of the following values of x satisfies the given equation?

A. -8

B. 4

C. 32

D. 128

Q10

GRID-IN ADVANCED MATH

$$\sqrt{x+4} = 11$$

What value of x satisfies the equation above?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.